

Lab Project:
Secure File transfer using protocol
SFTP.

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Secure File Transfer Using SFTP

Objective:

To securely transfer files between a Windows Server and client using SFTP (SSH File Transfer Protocol).

Prerequisites:

- A Windows machine with WinSCP installed.
- A Linux machine with an SSH server running.
- Network connectivity between the Windows and Linux machines.

Lab Setup:

1.Windows Server:

- Install WinSCP on windows from the following link.

<https://winscp.net/eng/download.php>

2.Linux Server:

- Install OpenSSH Server on Linux (If Not Installed)

`sudo apt update && sudo apt install openssh-server -y`

For Debian/Ubuntu

`sudo systemctl enable ssh && sudo systemctl start ssh`

Enable and start SSH

- Verify SSH is running:

`systemctl status ssh`

```
hadiya-zuberi@hadiya-zuberi-VMware-Virtual-Platform: ~  
service.  
Created symlink /etc/systemd/system/multi-user.target.wants/ssh.service → /usr/lib/systemd/system/ssh.service.  
hadiya-zuberi@hadiya-zuberi-VMware-Virtual-Platform:~$ sudo systemctl start ssh  
hadiya-zuberi@hadiya-zuberi-VMware-Virtual-Platform:~$ systemctl status ssh  
● ssh.service - OpenBSD Secure Shell server  
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enab>  
   Active: active (running) since Tue 2026-03-10 12:23:49 PKT; 6min ago  
TriggeredBy: ● ssh.socket  
   Docs: man:sshd(8)  
         man:sshd_config(5)  
  Process: 4929 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)  
 Main PID: 4930 (sshd)  
    Tasks: 1 (limit: 2220)  
  Memory: 1.2M (peak: 1.5M)  
     CPU: 140ms  
   CGroup: /system.slice/ssh.service  
           └─4930 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"  
  
Mar 10 12:23:49 hadiya-zuberi-VMware-Virtual-Platform systemd[1]: Starting ssh.>  
Mar 10 12:23:49 hadiya-zuberi-VMware-Virtual-Platform sshd[4930]: Server listen>  
Mar 10 12:23:49 hadiya-zuberi-VMware-Virtual-Platform sshd[4930]: Server listen>  
Mar 10 12:23:49 hadiya-zuberi-VMware-Virtual-Platform systemd[1]: Started ssh.s>  
lines 1-18/18 (END)
```

- Find the Linux machine's IP address:

`ip a | grep inet or ifconfig`

```
hadiya-zuberi@hadiya-zuberi-VMware-Virtual-Platform:~$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000  
    link/ether 00:0c:29:29:39:c6 brd ff:ff:ff:ff:ff:ff  
    altname enp2s1  
    inet 192.168.205.131/24 brd 192.168.205.255 scope global dynamic noprefixroute ens33  
        valid_lft 973sec preferred_lft 973sec  
    inet6 fe80::20c:29ff:fe29:39c6/64 scope link  
        valid_lft forever preferred_lft forever
```

Task:

(Before Proceeding create a text file in linux add some data in it or download anything from internet so that you can transfer it to the windows machine consider that file a top secret level document)

`nano secret_document.txt`

```
hadiya-zuberi@hadiya-zuberi-VMware-Virtual-Platform:~$ nano secret_document.txt
hadiya-zuberi@hadiya-zuberi-VMware-Virtual-Platform:~$
```

Add some content, e.g.:

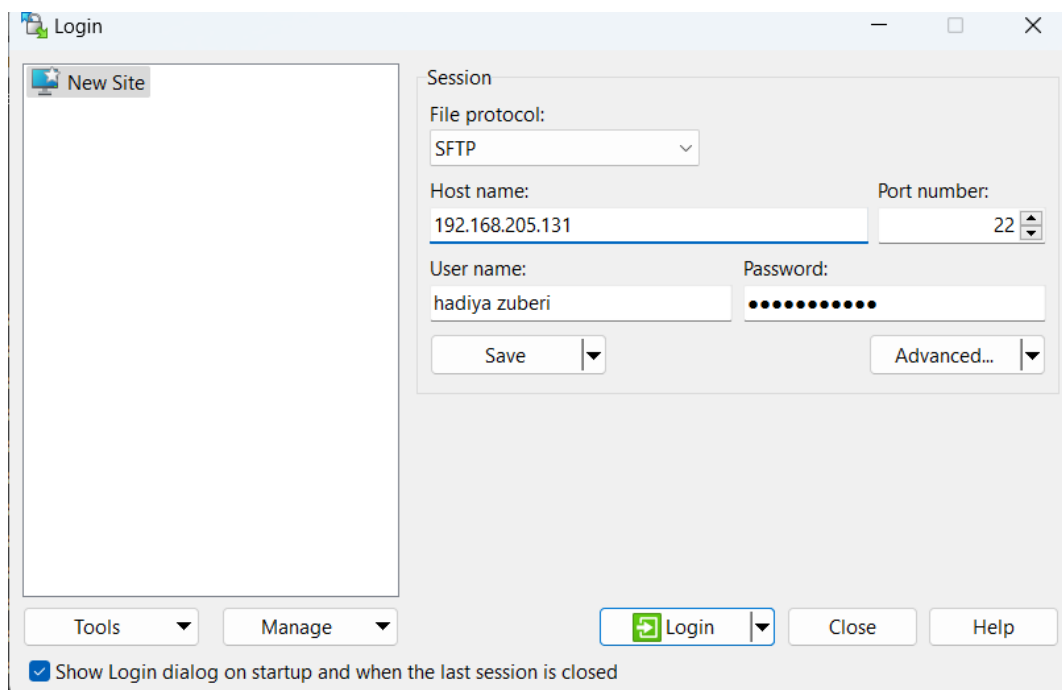
TOP SECRET - Lab Document

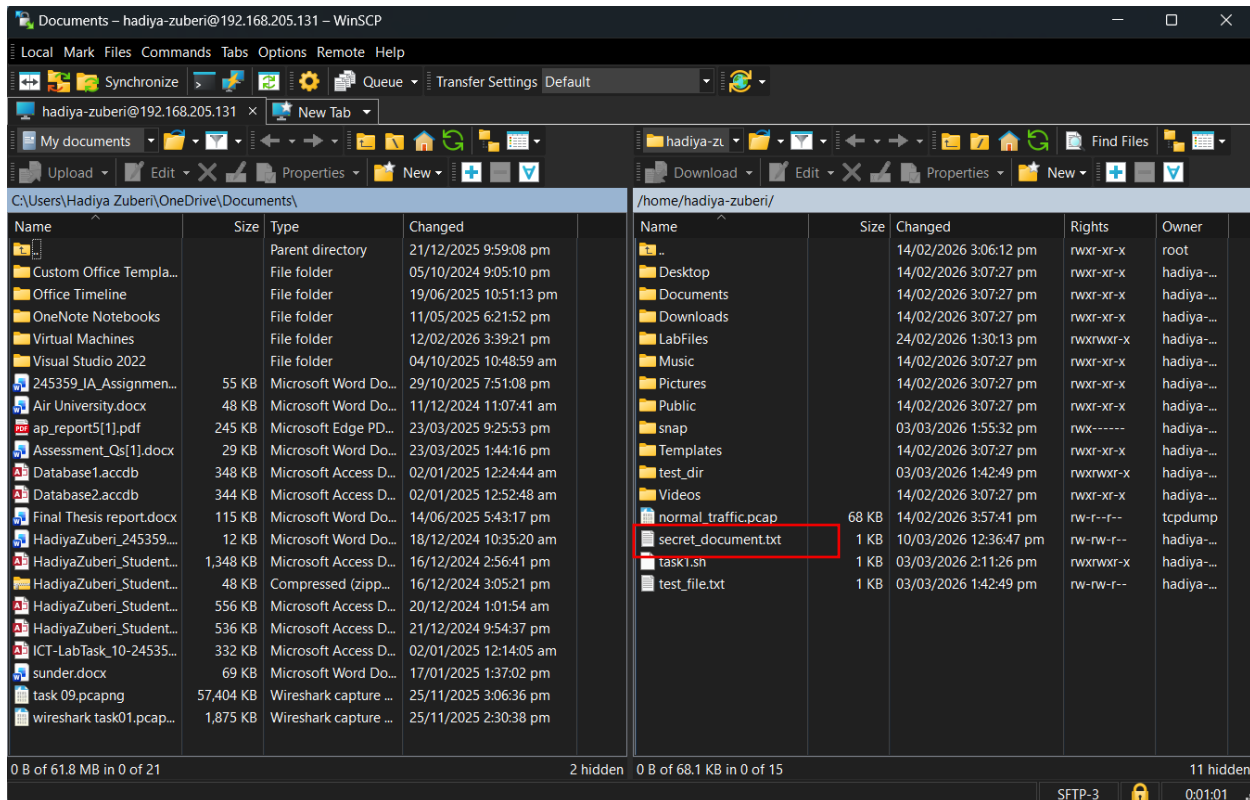
Project Code: ALPHA-7

Classified Data: 1234-ABCD

After adding the content save it with Ctrl + S and then exit using Ctrl + X

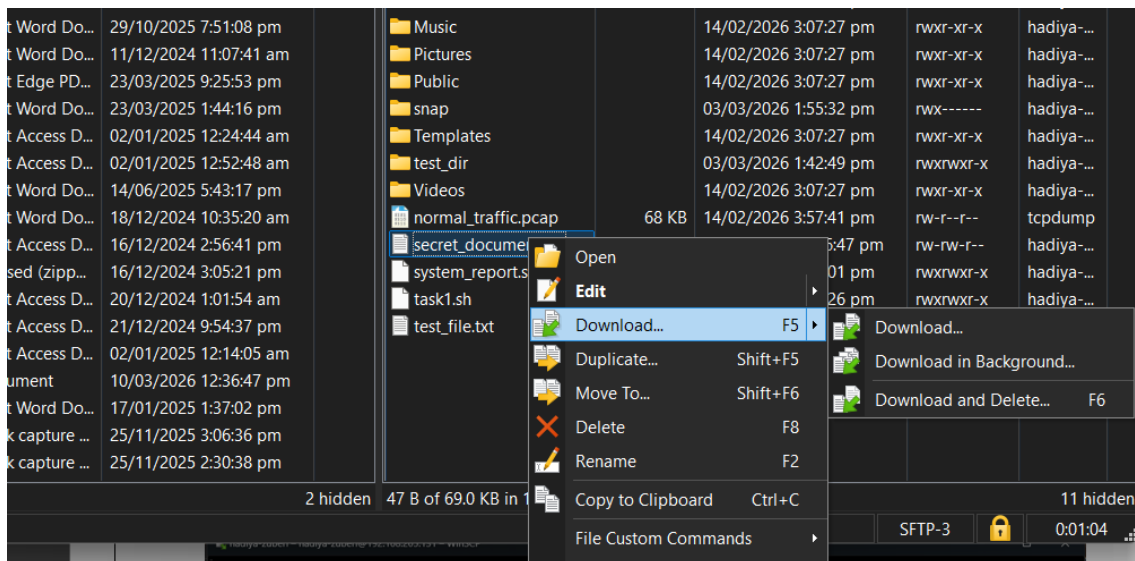
- Connect to Linux from Windows Using WinSCP
 - Open WinSCP.
 - Select SFTP as the file protocol.
 - Enter the Linux machine's IP address in the Host Name field.
 - Enter your Linux username and password.
 - Click Login to connect.

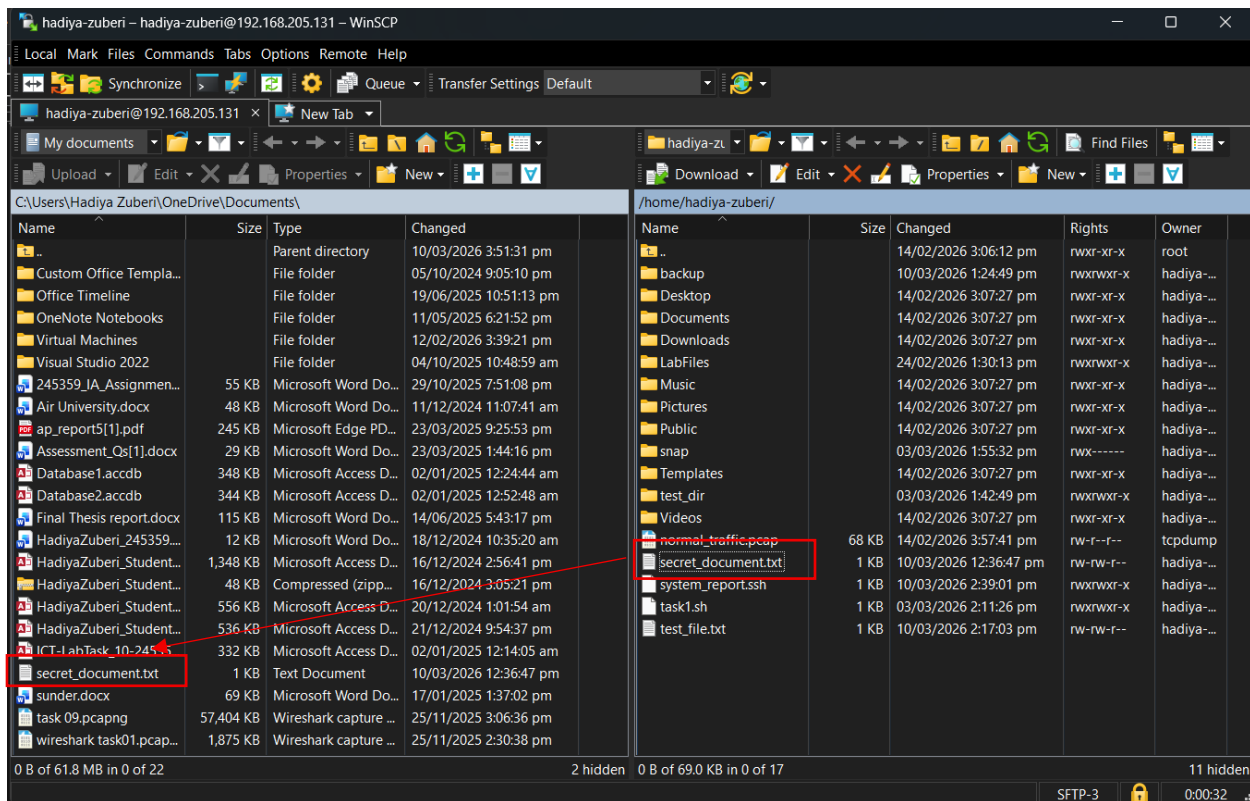




➤ Download Files from Linux to Windows

- Once connected, navigate to the file location on the Linux machine in the remote panel.
- Navigate to the desired location on Windows in the local panel.
- Drag and drop files from Linux to Windows to download them. OR right click the desired file and click download.





Note:

- Ensure the file appears in the specified Windows folder.
- Open the file to confirm it transferred correctly.

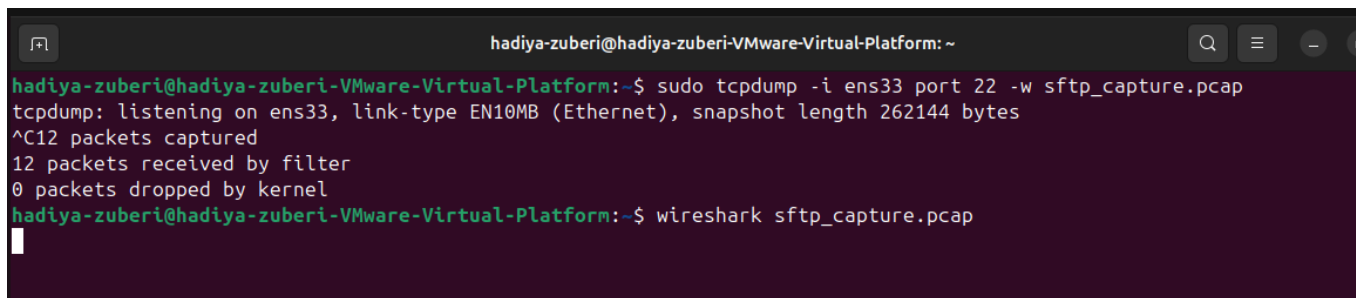
To prove the transfer was secure (encrypted):

1. Before transferring the file type this command in Ubuntu terminal to capture packets

`sudo tcpdump -i ens33 port 22 -w sftp_capture.pcap`

2. Transfer the file from WinSCP on windows. Go back to Ubuntu and press Ctrl + C to stop capture.
3. Open the pcap file in Wireshark

`Wireshark sftp_capture.pcap`



4. Filter port 22 to check whether the packets were encrypted or not.

Wireshark packet capture of an SFTP session. The filter is `tcp.port == 22`. The packet list shows several SSH packets (Client and Server) with encrypted data. The packet details for Frame 3 show the SSH protocol structure, including the '00 50 56' magic bytes.

No.	Source	Destination	Protocol	Length	Info
90	192.168.205.1	192.168.205.131	SSH	166	Client: Encrypted packet (len=166)
47	192.168.205.131	192.168.205.1	SSH	134	Server: Encrypted packet (len=134)
59	192.168.205.1	192.168.205.131	SSH	134	Client: Encrypted packet (len=134)
13	192.168.205.131	192.168.205.1	SSH	150	Server: Encrypted packet (len=150)
27	192.168.205.1	192.168.205.131	SSH	134	Client: Encrypted packet (len=134)
70	192.168.205.131	192.168.205.1	SSH	230	Server: Encrypted packet (len=230)
18	192.168.205.1	192.168.205.131	SSH	134	Client: Encrypted packet (len=134)
95	192.168.205.131	192.168.205.1	SSH	150	Server: Encrypted packet (len=150)
57	192.168.205.1	192.168.205.131	TCP	60	51615 → 22 [ACK] Seq=353 Ack=4
71	192.168.205.1	192.168.205.131	SSH	134	Client: Encrypted packet (len=134)
37	192.168.205.131	192.168.205.1	SSH	134	Server: Encrypted packet (len=134)
52	192.168.205.1	192.168.205.131	TCP	60	51615 → 22 [ACK] Seq=433 Ack=5

Frame 3: 134 bytes on wire (1072 bits), 134 bytes captured (1072 bits) on interface 0

Ethernet II, Src: VMware_c0:00:08 (00:50:56:00:00:08), Dst: 192.168.205.131

Internet Protocol Version 4, Src: 192.168.205.1, Dst: 192.168.205.131

Transmission Control Protocol, Src Port: 51615, Dst Port: 22

SSH Protocol

0000 00 0c 29 29 39 c6 00 50 56 c0 00 08 08
0010 00 78 6d d8 40 00 80 06 70 d1 c0 a8 cd
0020 cd 83 c9 9f 00 16 80 4f 43 9a cf e2 a4
0030 7f ff 2e 8d 00 00 c5 bd 10 5b 40 1d ff
0040 c6 fc 84 a7 a2 c6 12 2d 9d 32 bc 1f ce
0050 cd be dc a1 27 6b b3 38 a4 70 81 10 31
0060 12 45 1d f6 6b fc a8 12 42 07 39 7c a3
0070 7d 7a 60 5d fa 20 fa 5e ed 51 3a 8e 88
0080 77 29 a2 1c 5b 8f

Conclusion

This lab demonstrated key network security technique. A classified document was securely transferred from Ubuntu to Windows using SFTP over WinSCP, with Wireshark confirming the data was fully encrypted on port 22 and unreadable in transit. Both protocols together provide strong protection against unauthorized access and data interception.